Model selection:

For the implementation of my chatbot,i have used Meta Llama 3.2 1B Instruct model that is available through hugging face library.
The model is lightweight and instruction tuned.As compared to the base model ,instruction tuned model provides better responses which better follows the user prompt.
It easily integrates with the hugging face AutoTokenizer, AutoModelForCausalLM, and pipeline making it suitable for making better chatbot. To

The chatbot was built up using:
- Autotokenizer
- AutoModelForCasuallLM
- Pipeline

These components allowed the chatbot to interact with user while generating meaningful healthcare-related responses.

Prompt Engineering Techniques Used

1)ROLE PROMPTING:-

Through the help of role prompting, we assigned a role to the chatbot before answering the queries of the user input.
Providing a specific role to the chatbot help in generating output that iss more relevant and professional according to the healthcare.
Ex:-You are an experienced AI Healthcare Assistant helping doctors and patients.

2)FEW-SHOT PROMPTING:-

Through the help of Few_shot prompting,we provide several inputs or examples before asking the model to process new queries .
These examples helps the chatbot to analyze the examples through which it can generate the use queries same as the pattern of the above examples.

Ex:-Example 1:Input
Patient has cough and fever.
Output
Symptoms:
- Cough
- Fever
Severity:

3)COT PROMPTING:-

Through the hep of COT prompting ,the model analyze the patients symptoms carefully before providing the output of the final recommendations.
But according to the instruction we don't have to reveal the internal reasoning processes we only provide the final assessment and recommendations to the user.

RULES:
1)Analyze the symptoms of patient internally before answering it.
2)Do not reveal model internal reasoning process.
3)Answer only according to the user input.
4)If the input in not appropriate then ask for related question.
5)If the situation is crtical then suggest for immediate medical attention

CHALLENGES FACED:-

Model Selection

The first challenge was to select an appropriate model for the chatbot to work on.Because different models work in different parameters, so we required an instruction -tuned model for a better response.

PROMPTING

The second challenge was to design a better prompt for the model ,so that it can go through the prompt and provide a better response according to it.

Better Instruction

Another challenge was to provide better instruction to the model which will help in providing output that is more structured and reliable to the user input and can be able to provide better recommendations by following the instructions.

Improvements After Prompt Refinement:-

1)Chatbot provides better recommendations.

2)It provide the output in better structure format.

3)Provide the output in JSoN format.

4)Provide the summaries of the reports.

5)Provide different results according the role provided to the chatbot.